

Posterior Results

run_id 20260916_045356 | seed 20260914 | 2000 draws x 4 chains

1. Headline posterior

Parameter	mean	sd	r_hat	ess_bulk
e_R0	1.935	0.03202	1.001	7784
mea_R0	8.947	0.8771	1	8064
flu_R0	3.028	2.823	1.002	3478
n_R0	1.078	0.02139	1.001	2466
n_attack_rate	0.1409	0.03367	1.001	2466
log_intercept_e	-0.9979	0.08613	1	7335
e_alpha_nb	2.16	0.6253	1.001	7258
mea_alpha_nb	0.9538	0.318	1.001	7010
flu_alpha_nb	0.9213	0.3201	1.001	5612
class_re_sd	0.888	0.2519	1	2977
hyper_log_R0_mu	0.9376	0.3763	1.002	1608
hyper_log_R0_sd	0.7119	0.2221	1	3515

2. Identification check (raw variables vs Normal(0,1) prior)

Parameter	Posterior mean	Posterior sd	Updated by data?
e_log_R0_raw	-0.3747	0.4778	YES
mea_log_R0_raw	+1.6422	0.6227	YES
flu_log_R0_raw	+0.1327	0.5160	YES
n_log_R0_raw	-1.1481	0.5687	YES

A raw variable whose posterior still has mean 0 and sd 1 never entered the likelihood. This is the check that identified the v5 defect.

3. Posterior predictive checks

Disease	Coverage of 90% PI	Pred mean	Obs mean	Verdict
Ebola	95.7%	1.931	2.101	PASS
Measles	96.5%	2.167	2.186	PASS
Influenza	100.0%	135.883	108.500	PASS
Norovirus	100.0%	0.154	0.152	PASS

4. Residual serial correlation

Disease	lag-1 ACF	lag-2	Ljung-Box p	Verdict (rule $ r < 0.20$)
Ebola	+0.146	+0.155	0.041	PASS
Measles	+0.205	+0.148	0.050	structure remains
Influenza	+0.827	+0.403	0.002	structure remains

5. Final-size consistency

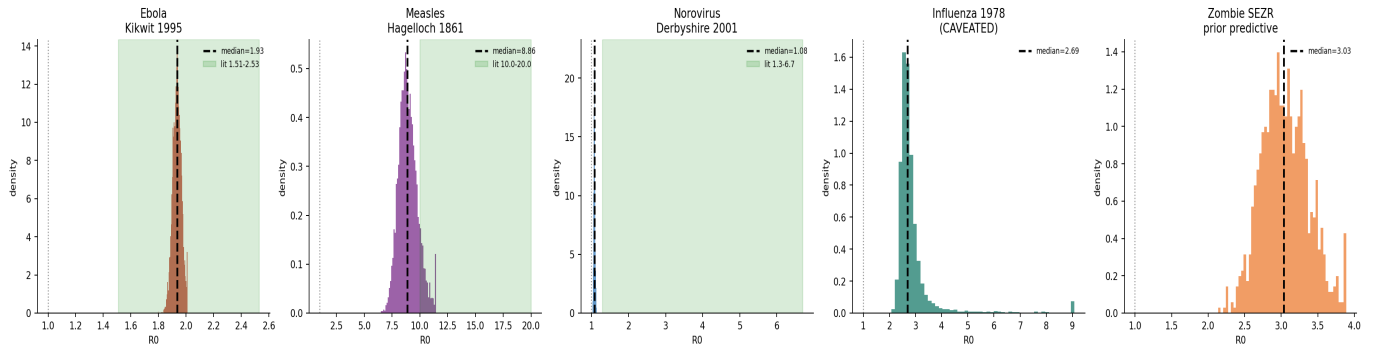
Disease	Implied AR	Reported AR	R0 from reported AR	Verdict
Influenza	91.4%	67.1%	1.66	CONFLICT
Norovirus	13.8%	15.2%	1.08	consistent

6. Prior dominance

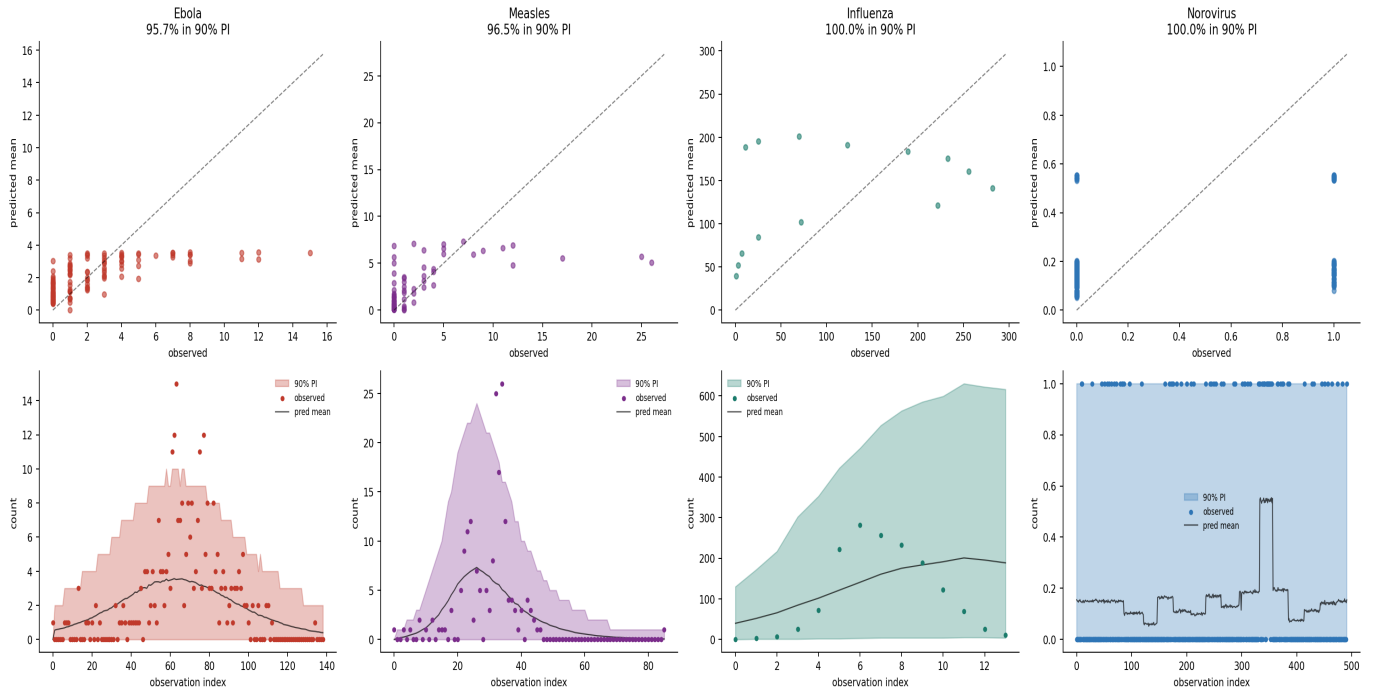
Parameter	Post/prior mean	sd ratio	Verdict
flu_alpha_nb	0.01	0.00	data-informed
e_alpha_nb	0.90	0.35	data-informed

7. Figures

BHMM posterior: R0 across diseases (v6 - R0 identified in every arm)



Posterior predictive checks



MCMC convergence

